Rahul Tandon

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Technical Skills

Languages: Python, Java, PHP, JavaScript, TypeScript, SQL
Frameworks and Libraries: React, Redux, FastAPI, Pandas, NumPy
AI Tools: Claude Code, OpenCode, Cursor, LangChain, RAG Pipelines, MCP Connectors, Structured Outputs, Agentic Frameworks
Databases: MySQL, PostgreSQL, Relational Database Design, Query Optimization, BigQuery
Monitoring and APIs: Grafana, Akamai, REST, GraphQL, gRPC, JSON, Kafka
DevOps and Infrastructure: AWS, Docker, Kubernetes, Jenkins, CI/CD, Linux, Distributed Systems, Cloud Native Architecture

Experience

Paycom Software, Inc. — **June 2023 - May 2026**
Software Developer — *Irving, Texas*

- Reduced content search latency by 55% across a learning platform serving 100,000+ concurrent users as measured by Grafana production dashboards by engineering a RAG pipeline with prompt engineering and LangChain orchestration using large language model APIs such as Claude and OpenAI, replacing keyword search with semantic retrieval backed by a PostgreSQL vector index, cutting average query time from 800ms to 360ms and delivering measurable performance improvements at scale.
- Reduced reporting turnaround by 60% as measured by stakeholder feedback cycles by co-owning dbt transformation models that moved raw operational data into BigQuery analytics-ready datasets, powering Grafana dashboards with real-time visibility across distributed production systems.
- Increased unit test coverage from 30% to 50% across Python backend services as measured by CI/CD pipeline reports by adopting Test-Driven Development practices within Agile sprint cycles, implementing JUnit-style TDD frameworks that reduced regression defects in production releases by 25% and improved overall code maintainability.
- Containerized and deployed production services using Docker in a Linux OS environment, establishing CI/CD pipelines that reduced manual release overhead and improved deployment reliability across engineering teams.
- Accelerated internal tool development velocity by 35% as measured by sprint delivery metrics by building scalable AI/ML infrastructure using MCP connectors and Claude Code, orchestrating multi-step language model agents with tool routing, credential management, and access control guardrails across distributed production systems.

Paycom Software, Inc. — **May 2022 - Aug 2022**
Software Development Intern — *Grapevine, Texas*

- Reduced API response times for high-traffic endpoints by 30% as measured by server profiling metrics by analyzing and optimizing execution paths within Python-based enterprise services and refactoring redundant query patterns.
- Delivered 2 production-ready RESTful and GraphQL API services in Python serving 20+ enterprise clients by designing standardized endpoints and containerizing deployments on Alpine Linux, enabling cross-platform portability across Unix environments.

C.T. Bauer College of Business, University of Houston — **Sept 2021 - Jan 2022**
Instructional Assistant — *Houston, Texas*

- Analyzed large consumer-producer datasets using Python (Pandas, NumPy) and SQL, generating actionable insights that supported student analytics projects on economic trends.
- Developed interpolation-based forecasting models in Python to evaluate and visualize market patterns, improving predictive accuracy of academic market simulation models by 15%.

UNBXD Software Private Ltd. — **Oct 2020 - Jul 2021**
Technical Support Engineer — *Bangalore, India*

- Analyzed large-scale product datasets using Python and SQL to optimize recommendation engine performance for enterprise e-commerce clients, improving click-through rates by 20% through data-driven merchandising decisions.
- Diagnosed data integrity and API integration issues across distributed client environments, collaborating with engineering teams to reduce production incidents and stabilize recommendation pipelines.

Projects

Video-Based Quiz Functionality - *Python, JavaScript, MySQL, Cloud File Share, Akamai*

- Led end-to-end development of a video-based assessment feature within the learning platform, adopted by 50+ clients and increasing quiz creation by 40%, strengthening client retention and upsell opportunities.
- Automated the video upload, watermarking, and encoding through integration with Cloud File Share, Akamai, and Encoding.com APIs - reducing manual intervention and upload errors by 90%.
- Designed persistent metadata tracking and playback reliability mechanisms, reducing support escalations by 80% and improving platform stability across large-scale deployments.

Sessions Management and Attendance System - *Python, JavaScript, MySQL*

- Reduced administrative overhead by 50% as measured by training manager feedback by architecting a state machine-driven sessions management system using Domain-Driven Design principles to model enrollment lifecycle transitions, supporting single-day and multi-day training programs with granular attendance tracking across partial participation scenarios.
- Designed and implemented asynchronous long-running processes (LRPs) workflows to handle high-volume batch operations (enroll, unenroll, attendance updates), ensuring system responsiveness, fault tolerance, and data consistency.
- Reduced engineering debugging time by 35% as measured by incident resolution reports by implementing a comprehensive audit logging framework that captured historical change events across the platform, enabling faster root cause analysis and improving data consistency across distributed enrollment workflows.

Education

University of Houston - Main Campus — **Aug 2021 - May 2023**
Master of Science in Computer Science - 3.8 GPA — *Houston, Texas, USA*

University of Delhi — **Jul 2016 - Jun 2019**
Bachelor of Science (Honours) in Computer Science - 8.3 GPA — *Delhi, India*